Supplement Table 4: Findings of studies examining the association between occupational sitting and heart diseases[1]

Cohort Study									
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country, cohort	n (% female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++), (+) (0)	Yes/no
1.	Allesøe 2014 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/ nursing	(1993–2008) 15 y	Self-reported OPA in 4 categories (questionnaire based on OPA assessment by Saltin & Grimby 1968), summed up into 3 categories: sedentary, moderate, high	IHD (national register)	Reference: moderate OPA sedentary OPA: HR=1.22 (0.94–1.59)[5] **Sedentary work not significantly associated with increased risk of IHD.** HR=3.31 (1.73–6.35) sedentary OPA/sedentary LTPA: HR=2.02 (1.28–3.76) sedentary OPA/vigorous LTPA **Sedentary work associated with increased risk of IHD only among nurses with sedentary and vigorous LTPA.**	++	Females only
.	Allesøe 2016 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/ nursing	(1993–2008) 15 y	S. above	Subgroup analyses for hypertensive subjects and IHD (national register)	Reference: normotensive-moderate OPA5 normotensive-sedentary OPA: HR=1.04 (0.77–1.42) hypertensive-sedentary OPA: HR=2.33 (1.48–3.67) **Combination of sedentary work and hypertension was associated with higher risk of IHD compared to normotensive nurses with moderate OPA, but not among normotensive nurses.**	++	Females only
3.	Ferrario 2018 Italy MONICA, PAMELA, and SEMM-Study	n=3574 (0 % F); 25–64 y; mean age: 40.8 y	Salaried employees (executive, white and blue collar workers) and general population (incl. working)	14 y median	Baecke Physical Activity Questionnaire (BPAQ): self-reported OPA categorized in tertiles into low, moderate or high OPA after occupation type	CHD (registries)	Reference: moderate/intermediate OPA low OPA HR=1.61 (1.02–2.52) **Low OPA showed a significant higher CHD risk.**	++	Males only
4.	Møller 2016 Denmark DWECS	1990: n=5420 (47.9 % F); 18–59 y	Working population	(1991–2010) mean 12.16 y; (2001–2010) mean 7.5 y	Self-reported work sitting time categorized as sedentary work >25 h/wk and non-sedentary work <25 h/wk	IHD (national register)	Reference: non-sedentary work RR=1 sedentary work (>25 h/wk spent sitting) RR=0.95 (0.78–1.16) Occ. sitting time per 10 h/wk increase: RR=0.98 (0.88–1.09) **No difference in risk of IHD was observed between sedentary and non-sedentary employees.**	++	No
5.	Johnsen 2016 Sweden WOLF study	n=9961 (31.2 % F); mean age: 42.7 y	Manual workers, low and interme-diate non-manual workers, profess-ionals	mean 13.1 y	Self-reported OPA levels categorized as:OPA 1: seated for more than 50 % of working day, no lifting or carrying; OPA 2: standing or walking, no lifting or carrying; OPA 3: lifting or carrying regardless of whether seated or standing/walking	MI (national and death register)	Reference: OPA 1 (50 % seated) OPA 2 (50 % standing+walking): HR=1.12 (0.80–1.56) **No significant associations between OPA and myocardial infarction.** M OPA 2: HR=1.11 (0.80–1.54) F OPA 2: HR=1.26 (0.52–3.03) **No major differences for OPA level and the risk of MI of HRs for men and women.**	+	Yes
6.	Smith 2018 Canada Canadian Community Health Survey (CCHS)	n=7320 (50 % F); 35–74 y	Different occupations general population (incl. working)	(1003–2015) 12 y	Job Exposure Matrix (JEM): primary type of posture for occupations in 4 categories after respondents occupation: primarily sitting (1), standing/walking (2); combination of sitting/standing/ walking (3); postures other than above named (4)	Incident heart disease (2 national databases on cases of heart diseases)	Reference: sitting occupations Sitting, standing, walking occupation: HR=0.97 (0.58–1.62)6 standing occupations: HR=2.18 (1.11–4.27) **Compared with occupations involving predominantly sitting, occupations involving predominantly sitting, standing, walking showed no significant different association, but standing occupations were associated with an approximately 2-fold risk of heart disease.** M sitting, standing, walking: HR=0.61 (0.33–1.13)6; F sitting, standing, walking: HR=1.80 (0.78–4.12)6 **Among men sitting, standing, walking showed a protective HR, but elevated HR among women.** HR estimates for occupations with sitting, standing, walking differed between men and women, but not reaching significance. HR estimates remained consistent among men and women for standing occupations: M standing occupations HR=2.01; F standing occupations HR=1.86	+	Yes

Case-Control Studies									
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of recruitment**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++), (+) (0)	Yes/no
1.	Cheng 2014 China INTER-HEART China study	Case: n=2909 (30.3 % F); mean-age: 55.2 y Control: n=2947 (30.5 % F); mean-age: 54,5 y 35–74 y hospital-based	General population (incl. working)	02/1999–03/2003	Self-rated OPA in 4 categories: mainly sedentary, predominantly walking at one level, mainly walking incl. walking up hill or lifting heavy objects, heavy physical labor	AMI (medical records)	Reference: mainly sedentary OPA walking at one level: OR=0.85 (0.70–1.05)[6] walking, climbing, lifting: OR=0.94 (0.70–1.27)[6] heavy physical labour: OR=1.37 (0.95–1.99)[6] **Association between levels of OPA and AMI risk factors were not significant, and not independently related to a reduced risk.** Sex-stratified results for interaction between OPA and AMI are shown in fig.1 without values: **Walking at work can reduce the risk of AMI: For women walking at one level had a significantly lower risk factor for AMI compared to mainly sedentary OPA. For men walking at one level showed a higher OR compared to mainly sedentary but not significant.**	+	Yes
2.	Held 2012 INTER-HEART study 52 countries worldwide	n=9805 case: n=14200 control: % F not reported hospital-based	General population (incl. working)	Elsewhere reported	Self-rated OPA in 4 categories: mainly sedentary, predominantly walking at one level, mainly walking incl. walking uphill or lifting heavy objects, heavy physical labor	MI (medical records)	Reference: sedentary OPA[6] walking incl. uphill and lifting: OR=0.89 (0.80–0.99)[6] walking at one level: OR=0.78 (0.71–0.86)[6] **Compared to sedentary OPA as reference, OR was reduced for walking at one level and for walking uphill and lifting objects.** walking at one level: OR=0.74 (0.65–0.83)[6] walking, climbing, lifting: OR=0.82 (0.71–0.95)[6] heavy OPA: OR=0.96 (0.80–1.16)[6] **Mild OPA appeared to be protective in middle- and high-income countries. Sex-stratified results were consistent with the overall analyses.**	+	Yes
3.	Ma2017 China	Case: n=354 (22.6 % F); mean age: 55.28 y; control: n=241 (45.2 % F); mean age: 51.94 % 24–65 y population-based	General population (incl. working)	12/2015–11/2016	Occupational physical activity questionnaire (OPAQ), self-rated OPA in 4 categories: sitting office work (sedentary OPA); mostly standing (light OPA); mostly walking (moderate OPA); heavy manual labor (heavy OPA)	CHD (medical records)	Reference: non-employed Sedentary occupation: OR=2.794 (1.526–5.115) **A sedentary occupation was a risk factor for CHD**. sedentary OPA: OR=2.794 (1.526–5.115) light OPA: OR=1.302 (0.695–2.439) moderate OPA: OR=0.972 (0.499–1.896) heavy OPA: OR=0.674 (0.336–1.352) **Significant linear relationship was observed between OPA and CHD: The lesser the OPA during working, the higher the incidence of CHD.** OR=2.213 (1.125–4.355) working 55 h/wk **Participants who worked prolonged hours (>40 h/wk) at sedentary occupations had an increased risk of CHD.**	+	No
4.	Selim 2013 Bangladesh	Case: n=200 (12.5 % F); mean age: 55.13 y; control: n=200 (14 % F); mean age: 48.91 y; 30–80 y hospital-based	General population (incl. working)	not reported	Sedentary occupation yes/no	ACS (national register)	OR=0.87 (0.429–1.762) (p=0.699)7 Amongst others sedentary occupation was found to be an independent predictor of acute coronary syndrome	0	No

Legend:
n (% F): number of participants (percent female; Female=F; Male=M; OPA: occupational physical activity
[1] In alphabetical order and after methodological assessment rating (high to low quality)
[2] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS)
[3] In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status if available, unless reported otherwise
[4] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
[5] Not adjusted for socioeconomic status
[6] Not adjusted for LPTA
[7] Adjustment not reported